

James Roggeveen, Ph.D.

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Agent Evaluation | Scientific Reasoning

SUMMARY

Researcher and applied mathematician building agent evaluations in scientific and engineering domains. Develops expert-designed tasks, programmatic graders, and evaluation infrastructure. Research background in differentiable scientific computing, inverse problems, and physical modeling.

RESEARCH EXPERIENCE

Google DeepMind Cambridge, MA | 2025–Present
Visiting Faculty Researcher

- Contributed scientific and engineering expertise to evaluating advanced AI reasoning, collaborating with researchers and domain experts to assess task quality and model capabilities.
- Publicly credited by Google DeepMind for foundational support to the [Gemini Deep Think team](#).

Harvard University Cambridge, MA | 2024–Present
Postdoctoral Fellow in Applied Mathematics | Advisor: Michael P. Brenner

- Built a cloud-hosted, end-to-end pipeline for model execution, collaborative testing, programmatic grading, and symbolic verification, allowing experts to focus on developing and refining domain-specific problems.
- Co-led **HARDMath2**, a 211-problem NeurIPS 2025 benchmark, and built its collaborative authoring and testing infrastructure, enabling 40+ graduate students to develop and peer-review original mathematics problems.
- Owned the evaluation infrastructure for **CMT-Benchmark**, enabling leading condensed-matter researchers to collaboratively collect, test, refine, and evaluate 50 original research-level problems across 17 frontier language models.
- Invented a meshless, differentiable spectral basis method for PDE inverse problems on irregular geometries; require coding agents to pass independent numerical checks before reporting results, helping catch invalid implementations while extending the solver to new applications.
- Designed a differentiable framework for identifying tensor-valued constitutive laws from sparse, partial flow measurements; manuscript under review at *Nature Communications*.

Princeton University Princeton, NJ | 2019–2024
Doctoral Researcher, Mechanical and Aerospace Engineering | Advisor: Howard A. Stone

- Derived models of complex fluid systems by applying mathematical theory, numerical simulation, and experiments, resulting in four first-author publications in fluid mechanics, soft matter, and biophysics.
- Created a statistical-inference pipeline for segmenting experimental data, validating model assumptions, and extracting material properties from noisy time series, reducing a roughly 30-minute analysis to seconds.

EDUCATION

Princeton University 2024
Ph.D., Mechanical Engineering

University of Cambridge 2019
Master of Advanced Study, Applied Mathematics | Distinction

Massachusetts Institute of Technology 2018
B.S., Mechanical Engineering | GPA: 5.0/5.0

SELECTED PUBLICATIONS AND BENCHMARKS

- J. V. Roggeveen*, E. Y. Wang*, *et al.* “HARDMath2: A Benchmark for Applied Mathematics Built by Students as Part of a Graduate Class.” *NeurIPS Datasets & Benchmarks*, 2025. [arXiv:2505.11774](#). *Equal contribution.
- H. Pan, J. V. Roggeveen, *et al.* “CMT-Benchmark: A Benchmark for Condensed Matter Theory Built by Expert Researchers.” *ICLR*, 2026. [arXiv:2510.05228](#).
- A. M. Sunol*, J. V. Roggeveen*, M. G. Alhashim, H. S. Bae, and M. P. Brenner. “Learning Constitutive Models and Rheology from Partial Flow Measurements Using Automatic Differentiation.” Under review at *Nature Communications*. [arXiv:2510.24673](#).
- J. V. Roggeveen and M. P. Brenner. “Meshless Solutions of PDE Inverse Problems on Irregular Geometries.” [arXiv:2510.25752](#), 2025.
- J. V. Roggeveen, H. Wang, Z. Shi, and H. A. Stone. “A Calibration-Free Model of Micropipette Aspiration for Measuring Properties of Protein Condensates.” *Biophysical Journal*, 2023. [DOI:10.1016/j.bpj.2023.09.018](#).

TECHNICAL EXPERTISE

Agent evaluation: Expert-driven benchmark development, task quality assessment, programmatic grading, symbolic and numerical verification, collaborative evaluation workflows

Scientific computing: Automatic differentiation, inverse problems, numerical PDEs, optimization, dynamical systems, statistical inference from experimental data, mathematical modeling

Programming and tooling: Python, JAX, SymPy, Mathematica, Google Cloud, Git